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SIMULATION EXPERIMENT – 8

Realization of FIR Filters Using Direct, Cascade and Linear-Phase Structures Using MATLAB MATLAB CODE:

```
clc;
clear;
close all;

%% FIR Filter
b = [1 2 1];
a = 1;

%% Input Signal
n = 0:50;
x = sin(0.2*pi*n);

%% ----------------------------
%% Direct Form
%% ----------------------------
y_direct = filter(b,a,x);

%% ----------------------------
%% Cascade Form
%% H(z)=(1+z^-1)(1+z^-1)
%% ----------------------------
b1 = [1 1];
b2 = [1 1];
y_temp = filter(b1,1,x);
y_cascade = filter(b2,1,y_temp);
```

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```
%% ----------------------------
%% Linear Phase Form
%% ----------------------------
if isequal(b,fliplr(b))
    disp('Linear Phase FIR Filter');
end
y_linear = filter(b,a,x);

%% ----------------------------
%% Impulse Response
%% ----------------------------
figure;
impz(b,a);
grid on;
title('Impulse Response');

%% ----------------------------
%% Magnitude and Phase Response
%% ----------------------------
figure;
freqz(b,a);

%% ----------------------------
%% Output Comparison
%% ----------------------------
figure;
plot(n,y_direct,'b','LineWidth',2);
hold on;
plot(n,y_cascade,'r--','LineWidth',2);
```

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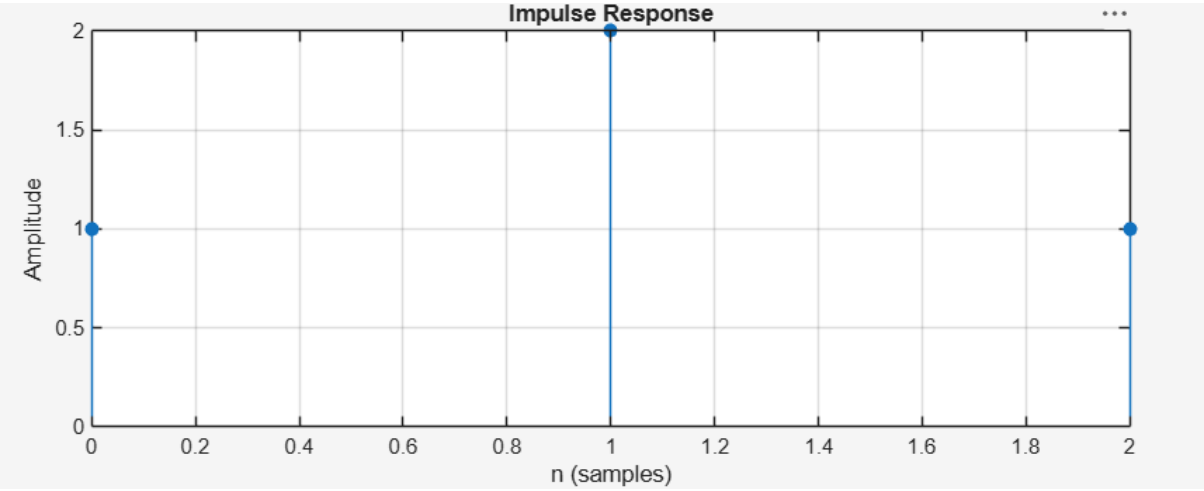
```
plot(n,y_linear,'k:','LineWidth',2);
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Direct vs Cascade vs Linear Phase');
legend('Direct','Cascade','Linear Phase');

%% ----------------------------
%% Verify Outputs
%% ----------------------------
disp('Maximum Error (Direct - Cascade)');
disp(max(abs(y_direct-y_cascade)));

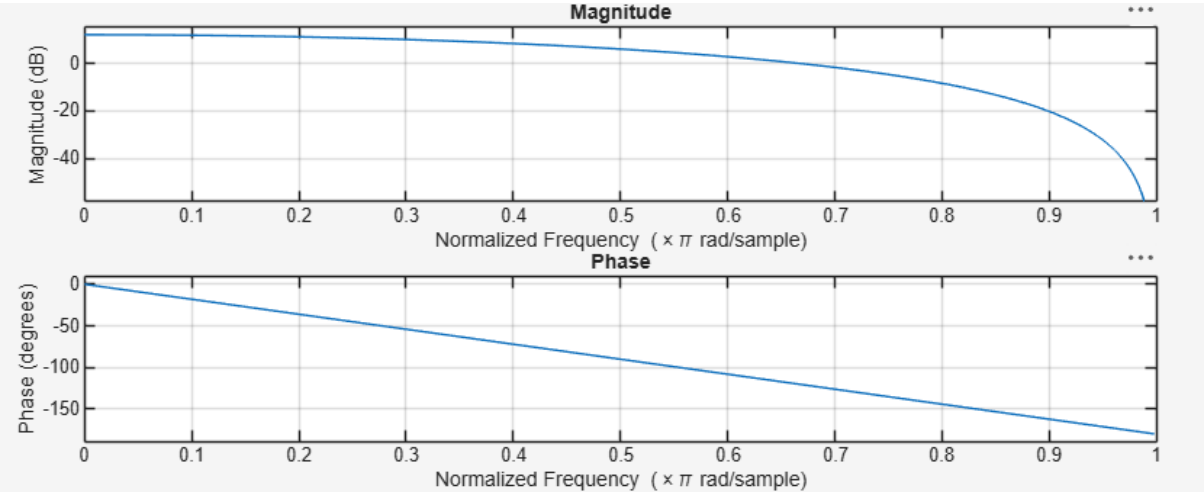
disp('Maximum Error (Direct - Linear)');
disp(max(abs(y_direct-y_linear)));

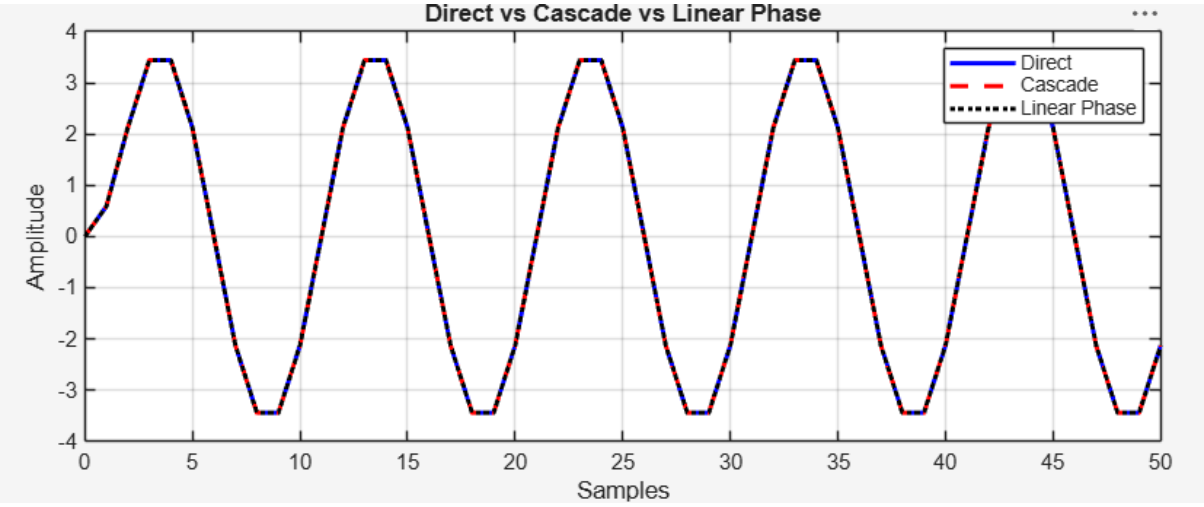
disp('FIR Coefficients');
disp(b);
```

OUTPUT



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```
Linear Phase FIR Filter
Maximum Error (Direct - Cascade)
   4.4409e-16

Maximum Error (Direct - Linear)
     0

FIR Coefficients
     1     2     1
```

RESULT

The FIR filter was successfully realized using **Direct**, **Cascade**, and **Linear-Phase** structures in MATLAB. The impulse response, magnitude response, phase response, and output response were obtained for all three realizations. The filter coefficients were found to be symmetric, confirming the **linear-phase property**. The outputs of the Direct, Cascade, and Linear-Phase realizations were identical, with **zero maximum error** between the Direct and Cascade forms as well as between the Direct and Linear-Phase forms, verifying that all three structures implement the **same FIR transfer function**. Thus, the objectives of the experiment were successfully achieved.